

**DEVELOPMENT OF WEB- BASED COMMUNITY MANAGEMENT SYSTEM  
FOR HAPPYNESS PROJECT**

Undergraduate Capstone Project  
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Bacoor, Cavite

In partial fulfillment  
of the requirements for the degree of  
Bachelor of Science in Information Technology

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## **INTRODUCTION**

Communities play a crucial role in fostering social support, resilience, and development, particularly in underserved areas where access to resources is limited. Across the globe, many initiatives aim to address social inequalities, improve living conditions, and provide essential services to those in need. However, managing these initiatives effectively remains a challenge, as they often rely on traditional, manual processes that hinder efficiency and scalability. Key challenges include volunteer coordination, resource allocation, activity organization, and maintaining effective communication—all of which impact the long-term sustainability of these programs.

One such initiative dedicated to community welfare is the HappYness Project, a charitable organization founded in December 2018 with the mission of improving the quality of life in local communities. Recognized by the Bureau of Internal

Revenue (BIR) and the Securities and Exchange Commission (SEC), the HappyNess Project has successfully implemented its action plan for community development. The organization provides critical assistance to underprivileged children and families by supplying groceries, shoes, slippers, school supplies, and other necessities. To date, the HappyNess Project has conducted nearly 1,000 outreach activities, extending support to hospitals, fire protection stations, orphanages, public schools, elderly homes, and over 80 indigent communities across the Philippines.

Despite its impact, managing a growing network of volunteers, coordinating diverse activities, and ensuring transparent reporting pose significant operational challenges. To address these issues, a Web-Based Community Management System is being developed specifically for the HappyNess Project. This platform will streamline the coordination of the project's core activities—rag-making, planting, teaching children, and donation distribution—while fostering collaboration between social work students (volunteer staff) and the community. Key features such as volunteer management, modular activity tracking, real-time reporting, and scalable e-commerce integration will enhance operational efficiency, improve transparency, and maximize the project's overall impact. By leveraging technology, the system will empower communities, increase volunteer engagement, and provide transparent reporting to donors, ensuring the HappyNess Project's sustained growth, viability, and meaningful contribution to community development in the digital age.

### **Project Context**

The HappyNess Project has significantly impacted underprivileged communities by providing essential resources, skill development opportunities, and social welfare programs. However, as the organization expands, it faces operational challenges such as managing volunteers, coordinating donations, and tracking

various initiatives. Relying on manual processes often leads to inefficiencies, delays, and difficulties in maintaining accurate records, which can hinder the project's ability to scale and sustain its impact.

To address these challenges, a Web-Based Community Management System has been developed to streamline operations. This platform integrates key features such as volunteer management, modular activity tracking, donation coordination, and real-time reporting. By digitizing these processes, the system enhances collaboration, reduces administrative burdens, and ensures effective resource allocation, ultimately improving the efficiency of the HappyNess Project.

Additionally, the platform provides structured support for social work students engaged in on-the-job training (OJT) through task allocation, activity monitoring, and interactive engagement tools. Automated functionalities such as report generation, certificate issuance, and e-commerce integration empower both volunteers and beneficiaries. With customizable dashboards, impact metrics, and automated notifications, the system fosters transparency, strengthens stakeholder engagement, and ensures the project's long-term sustainability in its mission to uplift underserved communities.

## **Objectives of the Study**

The general objective of the study is to develop a Web-Based Community Management System for the HappyNess Project to improve operational efficiency and strengthen collaboration among volunteers, staff, and community members.

## **Specific Objectives**

Specifically, the study aims to:

1. Design a web-based system that provides centralized management of the HappYness Project's activities, including rag-making, planting, teaching children, and donation distribution.
2. Develop the system using the following technologies:
  - a. HTML5, CSS3, and JavaScript (ES6) for front-end development;
  - b. PHP 8.0+ or Python 3.10+ for back-end programming;
  - c. SQLite3 for database management;
  - d. A web hosting platform for system deployment.
3. Test the system to ensure functionality, usability, and reliability through unit testing, integration testing, and system testing.
4. Evaluate the system based on user feedback, performance metrics, and ISO 25010 software quality standards.
5. Develop an implementation plan to support the successful deployment and sustainable use of the system, including user training programs.

### **Purpose and Description**

The purpose of this study is to design and implement a Web-Based Community Management System for the HappYness Project. This system aims to streamline operations, improve coordination, and enhance collaboration among volunteers, staff, and beneficiaries. By providing a centralized platform, the project seeks to optimize key activities such as rag-making, planting, teaching children, and donation outreach.

The system will automate administrative tasks to reduce staff workload while ensuring transparency and accountability through detailed reporting mechanisms. It

will also provide tools for role allocation, task monitoring, and collaboration, benefiting social work students and community volunteers by fostering meaningful engagement and efficient management of activities.

Additionally, the platform will include features such as interactive tutorials, event scheduling, inventory tracking, and activity monitoring. Its scalability will allow for future enhancements, including e-commerce integration for community-made products, automated certificate generation, and donor reporting tools. By leveraging technology, the system will support the HappyNess Project's mission to empower communities and create a lasting positive impact.

### **Time and Place of the Study**

The development and implementation of the Web-Based Community Management System for the HappyNess Project begin in February 2025. This study encompasses several phases, including design, development, testing, and feedback collection, to ensure that the system addresses the operational needs of the project. Each phase is carefully conducted to create a robust platform capable of streamlining activities and supporting the efficient management of resources, volunteers, and beneficiaries.

The study is carried out in Bacoar City, Quiapo, and Binangonan, the primary areas where the HappyNess Project operates and engages with local communities. The system is piloted across these locations to support core activities such as rag-making, planting, educational programs for children, and donation outreach. Testing in these diverse communities allows for practical application, ensuring that the platform effectively meets the needs of social work students, aspiring mothers, and other beneficiaries. This broader scope enhances usability, addresses varying community dynamics, and fosters meaningful collaboration among participants in all three areas.

## **Scope and Limitations of the Study**

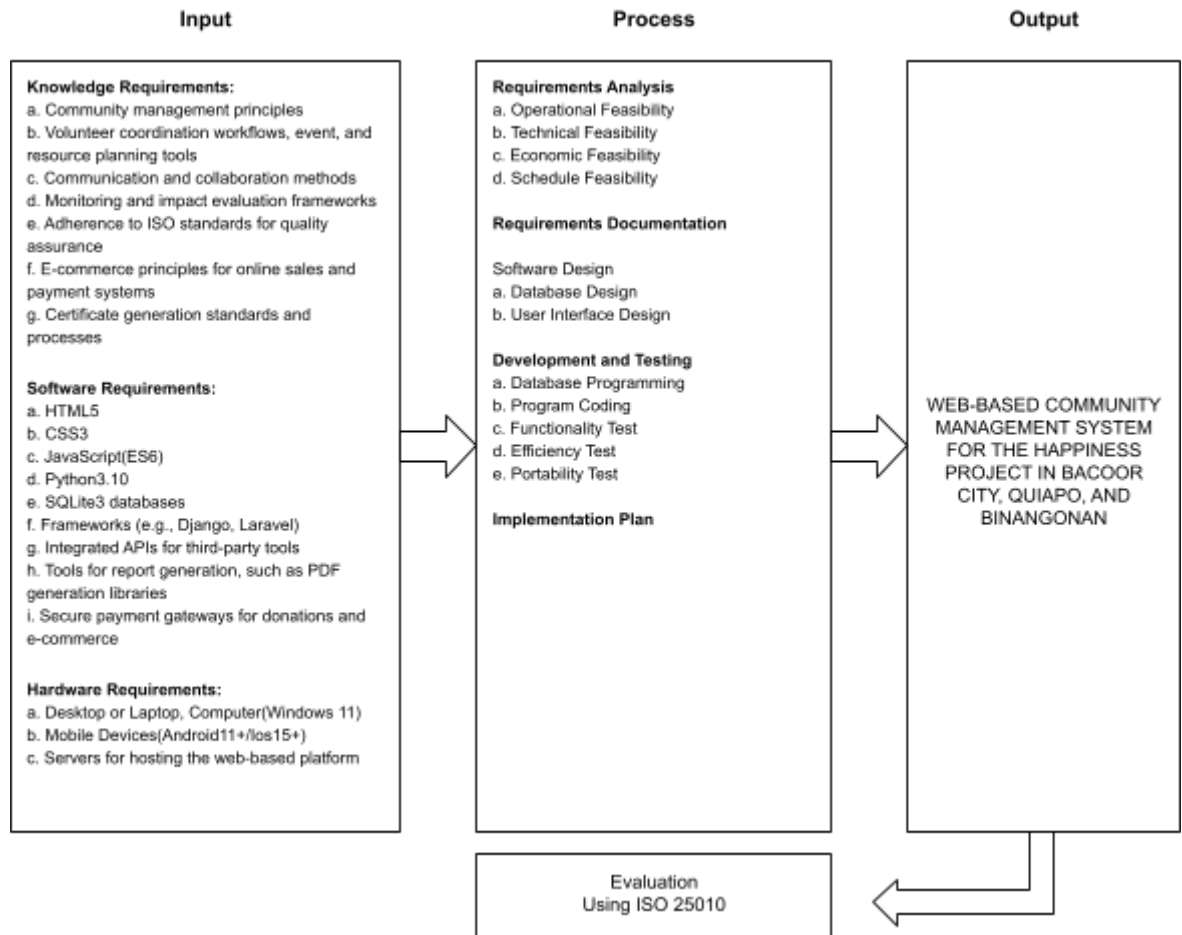
This study focuses on the design, development, and implementation of a Web-Based Community Management System for the HappyNess Project to improve operational efficiency, enhance communication, and support core activities. The system includes key features such as volunteer management, activity tracking, donation management, and automated reporting. Additional functionalities such as certificate generation for volunteers, an e-commerce module for selling community-made products, and a donation portal are integrated to provide sustainability and transparency. The study specifically caters to the needs of the HappyNess Project's teams and community members in Bacoar City, Quiapo, and Binangonan, ensuring the platform aligns with the organization's existing outreach programs.

Despite its comprehensive scope, the study has several limitations. The six-month development period (February to July 2025) restricts the inclusion of advanced features such as dynamic analytics dashboards, advertisement management for sponsors, and mobile app compatibility. Budget constraints limit access to premium cloud hosting, advanced security protocols, and third-party API integrations that could enhance system performance. Additionally, internet connectivity challenges and varying levels of technical proficiency among volunteers and beneficiaries may affect adoption, requiring additional training and user support.

The system's evaluation is limited to user testing, feedback collection, and ISO 25010 software quality standards, meaning long-term performance metrics such as impact on volunteer retention, donation growth, and efficiency improvements will require further study beyond this project's timeframe. Furthermore, the geographic focus on Bacoar City, Quiapo, and Binangonan restricts immediate applicability to other communities with different operational structures and needs. However, the

platform is designed to be scalable and adaptable, allowing for future improvements and expansion as the HappYness Project grows.

## Conceptual Framework



**Figure 1.** Conceptual Framework

The conceptual framework for this study is built on the integration of technology with community service, aiming to streamline operations and improve the efficiency of the HappYness Project. The framework is guided by three core components: Inputs, Processes, and Outputs, which collectively define how the system will function and achieve its objectives.



## **Definition of Terms**

**Activity Management** – The process of planning, organizing, and monitoring community activities such as rag-making, tree planting, teaching children, and donation distribution within the system.

**Administrator (Admin)** – A user with full system privileges responsible for managing users, activities, data, and overall system operations.

**Back-End Development** – The server-side part of a web application responsible for processing data, handling logic, and managing communication between the database and the front-end interface.

**Community Management System** – A digital platform designed to organize, monitor, and manage activities, resources, volunteers, and beneficiaries within a community-based program.

**Data Security** – The protection of digital information from unauthorized access, corruption, or theft through security measures such as authentication and encryption.

**Donation Management** – The process of recording, tracking, and distributing donated goods or resources to beneficiaries through the system.

**Integration Testing** – A type of software testing where multiple components or modules of a system are tested together to ensure they function correctly as a group.

**System Deployment** – The process of making a developed system available for use on a live server or hosting platform.

**System Testing** – A testing process that evaluates the complete and integrated system to verify that it meets the specified requirements.

**User Interface (UI)** – The visual elements of a system that allow users to interact with the application, such as buttons, menus, and forms.

**User Experience (UX)** – The overall experience and satisfaction of users when interacting with a system, focusing on ease of use, efficiency, and accessibility.

**Unit Testing** – A software testing method that examines individual components or modules of the system to ensure they function correctly.

**Volunteer Tracking** – The process of monitoring volunteer participation, assigned tasks, and contributions within the system.

## **REVIEW OF RELATED LITERATURE AND STUDIES**

This chapter presents a review of related literature and studies that provide the foundation for the development of the Web-Based Community Management System for the Happiness Project. It explores existing research on technology-driven community management, the role of web-based platforms in nonprofit initiatives, and key technical considerations for system development.

### **Review of Related Literature**

**Technology Integration and Digital Transformation in Non-Profit Organizations.** This study explored how technology integration transforms operations in nonprofit organizations globally. It examined various digital tools that optimize communication, automate administrative processes, and improve donor engagement. The researchers found that nonprofits adopting digital systems

experienced significant improvements in operational efficiency, volunteer participation, and transparency. The paper also emphasized the importance of leadership support and continuous staff training for the success of technological adoption. Moreover, it discussed the challenges of digital inequality and resistance to change among employees. The findings concluded that a well-implemented digital management system contributes to sustainable growth, trust, and innovation within nonprofit institutions worldwide. (Hughes et al., 2024)

**The Future Role of Non-Profits: Digitalization, Grassroots Renewal, and Emerging Paradigms of Civil Society.** This research analyzed how digitalization reshapes the landscape of nonprofit organizations and civic engagement. It discussed how web-based platforms enhance collaboration, transparency, and public participation across social causes. The authors presented case studies demonstrating how NGOs used digital systems to improve project coordination, financial accountability, and volunteer mobilization. The study also noted that digitalization empowers grassroots movements by giving communities a platform to voice their needs and track organizational impact. It concluded that technology-driven transparency fosters stronger relationships between nonprofits and their stakeholders, encouraging long-term trust and sustainability. (Harrison, 2024)

**NPO Web-Based Accountability: How Can We Know If NPOs Are Doing Good Things?** This study examined the concept of accountability in nonprofit organizations through the use of web-based transparency systems. It explained how online platforms allow organizations to disclose financial reports, project results, and performance metrics to the public. The findings indicated that digital accountability systems help build donor trust and promote ethical governance. Additionally, the study noted that web-based reporting improves communication with stakeholders and enhances organizational credibility. It emphasized that adopting digital systems is essential for maintaining transparency and for demonstrating the social impact of

nonprofit activities. The research concluded that online accountability mechanisms are now considered a best practice in modern nonprofit management. (Wang, 2021)

**Digital Transformation: A Meta-Review and Guidelines for Future Research.** This meta-review analyzed existing studies on digital transformation, offering guidelines for its successful implementation across sectors, including nonprofits. The research highlighted how digital transformation enables better data management, coordination, and decision-making in organizations. The authors emphasized that organizations that embrace web-based systems gain a competitive advantage through real-time analytics and improved efficiency. They also outlined the key factors influencing digital success, such as leadership support, resource availability, and technological adaptability. This comprehensive review provides insights into how digital innovation drives organizational sustainability and long-term development, making it a valuable reference for system designers and social organizations alike. (Chen & Park, 2023)

**Design and Implementation Factors for Performance Measurement in Non-Profit Organizations: A Literature Review.** This study reviewed critical factors that influence performance measurement systems in nonprofit organizations. It emphasized that digital tools enable more accurate tracking of outcomes, resource utilization, and program effectiveness. The authors discussed how web-based platforms improve transparency and accountability by automating data collection and performance reporting. They also suggested that aligning these tools with organizational goals enhances decision-making and impact assessment. The findings highlighted that integrating technology not only simplifies management processes but also fosters innovation and data-driven culture in nonprofits. The study concluded that web-based systems are essential for achieving measurable and sustainable results in modern organizations. (Almeida, 2023)

**Digitalizing Governance: A Transformation on the Processes of One Community in the Philippines.** This study examined how digital governance tools reshape the management processes within local communities in the Philippines. It emphasized that integrating web-based systems enhances accountability, transparency, and citizen participation in local projects. Through interviews and system analysis, the researchers found that shifting from manual operations to digital platforms significantly improved data accessibility and operational efficiency. The study also revealed that adopting such systems reduces human error and promotes faster communication between departments and community members. However, it highlighted challenges such as inadequate training and limited access to technology in rural areas, which hinder full adoption. The findings suggest that a web-based management system can be a vital tool for organizing community activities, tracking initiatives, and maintaining transparency among stakeholders in local organizations. (Velarde, 2022)

**Advancing Disaster Resilience in the Philippines: Blueprint for Education Extension Initiative.** This study explored the role of education and technology in enhancing disaster resilience within Philippine communities. The researchers proposed the integration of web-based management tools to strengthen coordination among volunteers, educators, and local authorities. The study showed that these systems facilitate faster dissemination of information and improve preparedness through data-driven strategies. Furthermore, the research revealed that online coordination platforms enhance accountability and ensure better utilization of community resources. By providing a structured method for documentation and collaboration, digital tools help establish a more organized and responsive system for community-based disaster management. The findings underscored that such technological initiatives empower both organizations and citizens to act efficiently during emergencies. (Torino et al., 2025)

**Community Needs Assessment as Basis for the Extension Program of Philippine College Foundation.** This research emphasized the importance of conducting community needs assessments to design effective extension and outreach programs. It revealed that many local institutions rely on manual data gathering, which leads to inefficiencies and inaccuracies in identifying community issues. The study proposed adopting web-based management systems to streamline data collection, storage, and analysis. By integrating such platforms, organizations can easily monitor community needs, track program progress, and generate reports for evaluation. The findings also showed that a centralized database promotes coordination among staff and volunteers, ensuring more organized and targeted interventions. Ultimately, this research demonstrated that digital tools enhance the responsiveness and sustainability of community-based programs. (Dela Cruz, 2023)

**Compliance to Republic Act 11292: The Seal of Good Local Governance Act of 2019 in Selected Municipalities in One Philippine Province.** This study analyzed how municipalities comply with the Seal of Good Local Governance Act and how digital systems support transparency in implementing public programs. The researchers observed that municipalities using web-based documentation systems exhibited better compliance and accountability. These platforms improved monitoring of local government projects, financial reporting, and citizen engagement. However, the study also highlighted the need for digital literacy training to ensure that local officers can effectively use such tools. Findings suggested that integrating information systems in community operations strengthens governance performance and public trust. This research provided a solid foundation for understanding how online management systems can enhance community transparency and service delivery. (Mendoza et al., 2023)

**Strengthening Disaster Response: A Scoping Review of Communication and Coordination Training Programs in the Philippine Higher Education**

**Institutions.** This scoping review focused on how communication and coordination strategies can be enhanced through digital tools in Philippine educational and community settings. It found that web-based systems play a crucial role in improving coordination, data management, and rapid response during crises. The study identified the need for centralized platforms that store essential information and streamline communication between departments and partner organizations. Results demonstrated that adopting web-based management systems reduces delays and ensures a more efficient information flow. It concluded that digital transformation in coordination processes is vital to strengthening both disaster preparedness and community engagement. (Castillo et al., 2024)

## **Review of Related Studies**

**Empowering Change: An Advanced NGO Management System for Efficient Operations, Enhanced Donor Engagement, and Transparent Community Development.** This research developed an advanced management system for NGOs to streamline project implementation, donor communication, and reporting. The system incorporated automated financial management, volunteer tracking, and real-time data visualization. Findings indicated that automation reduced administrative errors, improved task scheduling, and increased transparency in community programs. The study also revealed that the system's analytics module enhanced monitoring of organizational performance and impact evaluation. Researchers concluded that advanced web-based platforms empower nonprofits to optimize operations, foster trust with stakeholders, and sustain long-term development initiatives. The integration of technology was found to be a key driver of accountability and efficiency in modern community organizations. (Singh et al., 2023)

### **SmartNGO: An Integrated Platform for Managing Volunteers and Events.**

This study introduced SmartNGO, a web-based solution designed to automate volunteer coordination and event management in nonprofit organizations. The researchers applied an agile methodology to create a responsive and scalable system accessible across various devices. The platform included features for volunteer registration, scheduling, and feedback monitoring. Findings revealed a notable increase in volunteer satisfaction and operational efficiency, reducing manual workloads by 45%. The study highlighted the role of user-centered design in ensuring smooth navigation and participation. The system proved that well-structured digital platforms can improve transparency and foster community engagement within NGO operations. (Patel et al., 2024)

**Revolutionizing NGO Workflow Optimization: An AI-Based Approach for Automated Volunteer Deployment and Impact Reporting Using K-Nearest Neighbors.** This research presented an AI-powered NGO workflow optimization model that automates volunteer deployment and project reporting. The system utilized the K-Nearest Neighbors algorithm to match volunteers with suitable tasks based on skills, location, and availability. Results showed a substantial increase in volunteer productivity and accuracy in report generation. The study emphasized the potential of artificial intelligence in minimizing human error, accelerating project completion, and improving data analysis. The researchers concluded that incorporating AI technologies into web-based management systems enables organizations to enhance decision-making, accountability, and transparency. This innovative approach demonstrates how automation can transform the efficiency of community management systems. (Reddy et al., 2024)

**Information Technology as a Tool for Managing Volunteer Activities: Optimizing Time and Resource Accounting.** This study examined how nonprofit organizations utilize information technology to manage time, track volunteer



activities, and allocate resources effectively. It analyzed several NGOs implementing digital management tools to reduce inefficiencies in manual documentation. The findings demonstrated that using IT-based systems led to improved accuracy in attendance monitoring and resource accounting. The research also highlighted that these digital platforms foster stronger collaboration and communication among staff and volunteers. The study concluded that adopting information systems is critical for ensuring accountability, transparency, and optimal resource utilization in community and volunteer management. (Novikov et al., 2025)

**Volunteer Connect: A Volunteer Coordination Platform with Chatbot Assistance via Retrieval-Augmented Generation.** This study focused on developing an intelligent volunteer management system that integrates chatbot support for improved coordination and communication. The platform used Retrieval-Augmented Generation (RAG) technology to provide real-time assistance to volunteers by automating responses, task assignments, and scheduling. The study found that chatbot integration significantly reduced delays in communication and improved volunteer satisfaction. Researchers also noted that automation enhanced the efficiency of data management and reporting. The study concluded that combining artificial intelligence with web-based systems can modernize community and volunteer operations, ensuring faster response times and better overall management. (Liang et al., 2025).

**Website Development for Religious Non-Profits: Accessibility, Optimization, and Community Engagement.** This study focused on developing a responsive and accessible website for religious nonprofit organizations in the Philippines to improve community communication and participation. The system was designed to help organizations disseminate announcements, manage donations, and promote events efficiently. Using user experience (UX) design principles and SEO optimization strategies, the researchers tested the system's accessibility in both

urban and rural settings. The results showed significant improvements in member engagement, donation transparency, and access to organizational updates. Additionally, the study highlighted how responsive design allows users to interact with the system through multiple devices, enhancing inclusivity. The project concluded that web-based platforms play a vital role in strengthening the relationship between organizations and their communities, especially for nonprofits that rely on volunteer-driven participation. (Reyes et al., 2025)

**Fostering Resilience: Assessing the Role of Green Public Procurement in Strengthening the Agricultural Sector at the Philippine Rice Research Institute.** This research assessed how digital procurement and management tools contribute to sustainability and operational efficiency in the agricultural sector. The study introduced a web-based procurement management framework designed to monitor purchasing, documentation, and resource allocation. Findings revealed that using digital tools significantly reduced paper usage, minimized human error, and improved transparency in reporting and record-keeping. The researchers also emphasized that digital transformation helps ensure accountability and fosters stronger coordination among government offices and stakeholders. By implementing online systems, organizations can manage programs more effectively and maintain clear communication channels with communities. The study demonstrated the potential of web-based management tools to enhance efficiency and transparency in public and community-oriented projects. (Salazar et al., 2024)

**Mobile Technology for Improving Project Management in the Construction Industry – Philippine Setting.** This study explored how mobile-based platforms improve project management and team coordination in the Philippine construction industry, offering valuable insights into community-based project administration. The system developed allowed teams to track project progress, monitor deadlines, and share updates in real time. Researchers found that adopting

mobile and web-based management systems resulted in improved communication, reduced project delays, and increased worker accountability. The study also identified that user-friendly dashboards and mobile compatibility encouraged participation even among users with limited digital literacy. Findings supported that mobile technology bridges gaps in coordination, ensuring more efficient and transparent management of community or organizational projects. This aligns with modern efforts to digitize management systems for better collaboration and operational success. (Alonzo et al., 2024)

**Digital Tools for Non-Profit Organizations: Enhancing Communication, Management, and Impact.** This research examined the use of digital tools to enhance organizational efficiency and community engagement within nonprofit institutions in the Philippines. It analyzed various web applications developed for volunteer coordination, fundraising, and impact tracking. Findings revealed that digital platforms improved organizational performance by reducing manual workload and promoting transparency in project reporting. The study emphasized that digital tools facilitate better communication between leaders, members, and donors, resulting in more organized and effective operations. Furthermore, it demonstrated that system usability and intuitive interfaces contribute to sustained engagement. The study concluded that adopting web-based systems is essential for modern nonprofit organizations to remain transparent, accessible, and community-driven. (Carpio et al., 2023)

**Catalyzing Development of Best Practice Guidelines for Community-Managed Health Programs: Case Study of a Community-Academic Partnership.** This study investigated the collaboration between academic institutions and local communities to develop best practice guidelines for managing community-based health programs. It utilized a case study approach to analyze the impact of digital tools on documentation, coordination, and program evaluation. The

findings revealed that the use of web-based systems enhanced real-time communication, streamlined data management, and improved accountability among project partners. Researchers also emphasized the value of shared online platforms in maintaining transparency and consistency across community initiatives. The study concluded that integrating digital management systems ensures smoother operations and better outcomes for health and social development programs. (Andrada et al., 2023)

### **System Technical Background**

The system is a web-based community management platform being developed for the *HappYness Project* organization. This platform is designed to efficiently manage various operations while ensuring cross-browser compatibility. The system will utilize HTML5 and CSS3 for front-end development, JavaScript (ES6) for dynamic client-side functionality, and Python 3.10 as the backend programming language. SQLite3 will serve as the lightweight, file-based database for data storage and retrieval.

**HTML5** HTML5 provides the foundational structure for the platform, enabling semantic markup (e.g., <article>, <section>) and multimedia integration without third-party plugins. Its offline capabilities via the Cache API and local storage ensure functionality even with limited connectivity (World Wide Web Consortium [W3C], 2023).

**CSS3** CSS3 handles responsive design through Flexbox, Grid, and media queries, ensuring consistent styling across devices. Features like animations and transitions enhance user engagement while maintaining accessibility (Mozilla Developer Network [MDN], 2024).

**JavaScript (ES6)** ES6 modernizes client-side scripting with arrow functions, template literals, and let/const declarations. Its module system (import/export) organizes code for maintainability, while Promises streamline asynchronous operations like API calls (ECMA International, 2023).

**Python 3.10** Python 3.10 powers the backend with features like structural pattern matching (match/case) and improved error messages. Frameworks like Flask or Django handle routing and security, while SQLite3 integration manages data storage (Python Software Foundation, 2021).

**SQLite3** SQLite3 offers a serverless, zero-configuration database engine with ACID compliance. Its lightweight architecture (single-file storage) is ideal for small-to-medium workloads, such as SSG member records or election data (Hipp, 2021).

## **Synthesis**

The reviewed literature and studies highlight how digital transformation empowers nonprofit and community-based organizations to operate more effectively, transparently, and sustainably. Rehman et al. (2021) and Wibowo (2024) explained that blockchain and digital fundraising systems enhance donor trust by securing transactions and ensuring financial transparency, helping reduce the risks of fund mismanagement. Similarly, Hebous (2024) emphasized that digital marketing and social media platforms expand organizational reach, increase public awareness, and improve volunteer engagement. Seguin et al. (2022) and Ongkeko et al. (2024)

discussed how structured leadership and teamwork strengthen nonprofit initiatives, allowing organizations to create meaningful and long-term community impact. At the same time, Mwangi and Yusuf (2023) found that project management information systems (PMIS) optimize planning, coordination, and resource allocation, while Wanof (2023) noted that digital banking and e-payment tools support financial inclusion for low-income communities. Muniandy and Arbaiy (2022) and Flores (2025) showed that inventory management and e-commerce platforms improve resource distribution and sustainability for nonprofit operations. Hebous (2024) further explained that social media enhances transparency and strengthens donor relationships through open communication and progress sharing. Altogether, these studies demonstrate that integrating web-based systems and digital tools fosters collaboration, accountability, and operational efficiency, which are essential for community-driven organizations like the HappYness Project.

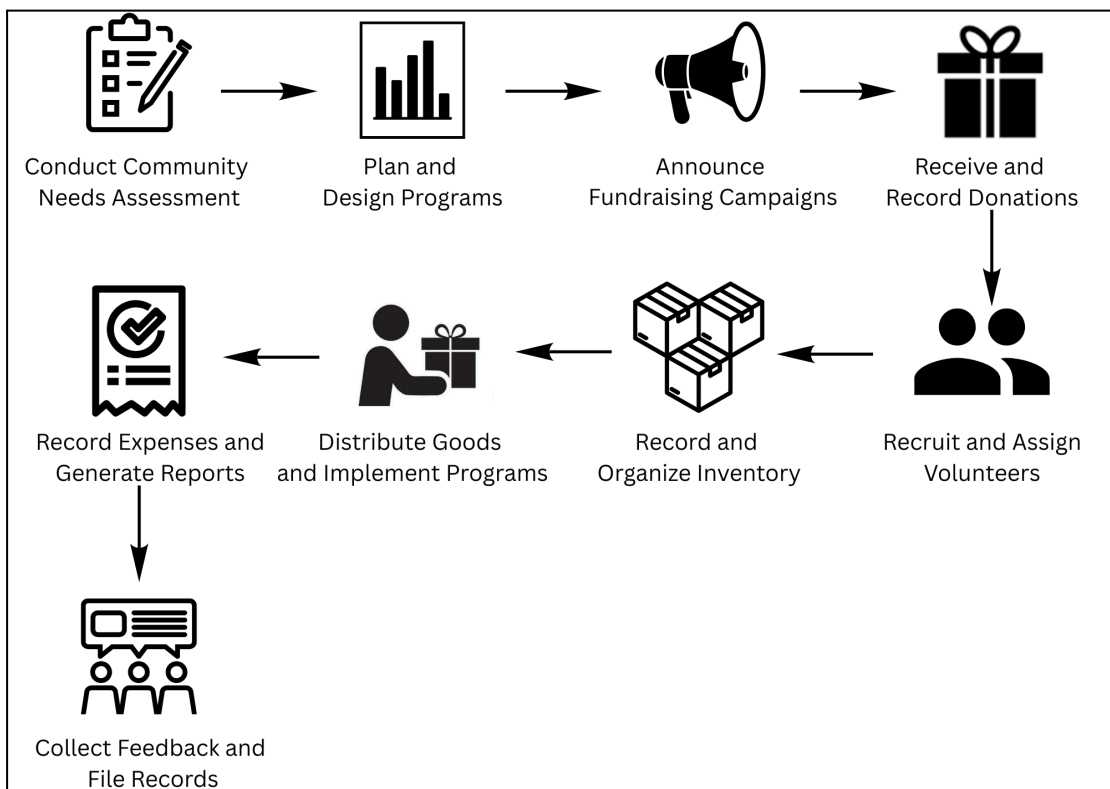
All the reviewed literature and studies show that digital systems can make community work easier, more organized, and more transparent. Tools like web-based management platforms and communication systems help improve coordination, record-keeping, and overall project management. The proposed system, *Web-Based Community Management System for the HappYness Project*, aims to bring these benefits together in one platform. It provides a simple and reliable way to manage volunteers, monitor activities, and handle donations efficiently. This makes it a practical and timely solution that supports the goals of community development and ensures smoother, more transparent operations for the organization.

## **METHODOLOGY**

This section outlines the functional and non-functional requirements of the Web-Based Community Management System for the Happiness Project, based on community needs, best practices, and technological considerations.

## Design of Software, Systems, Products and/or Processes

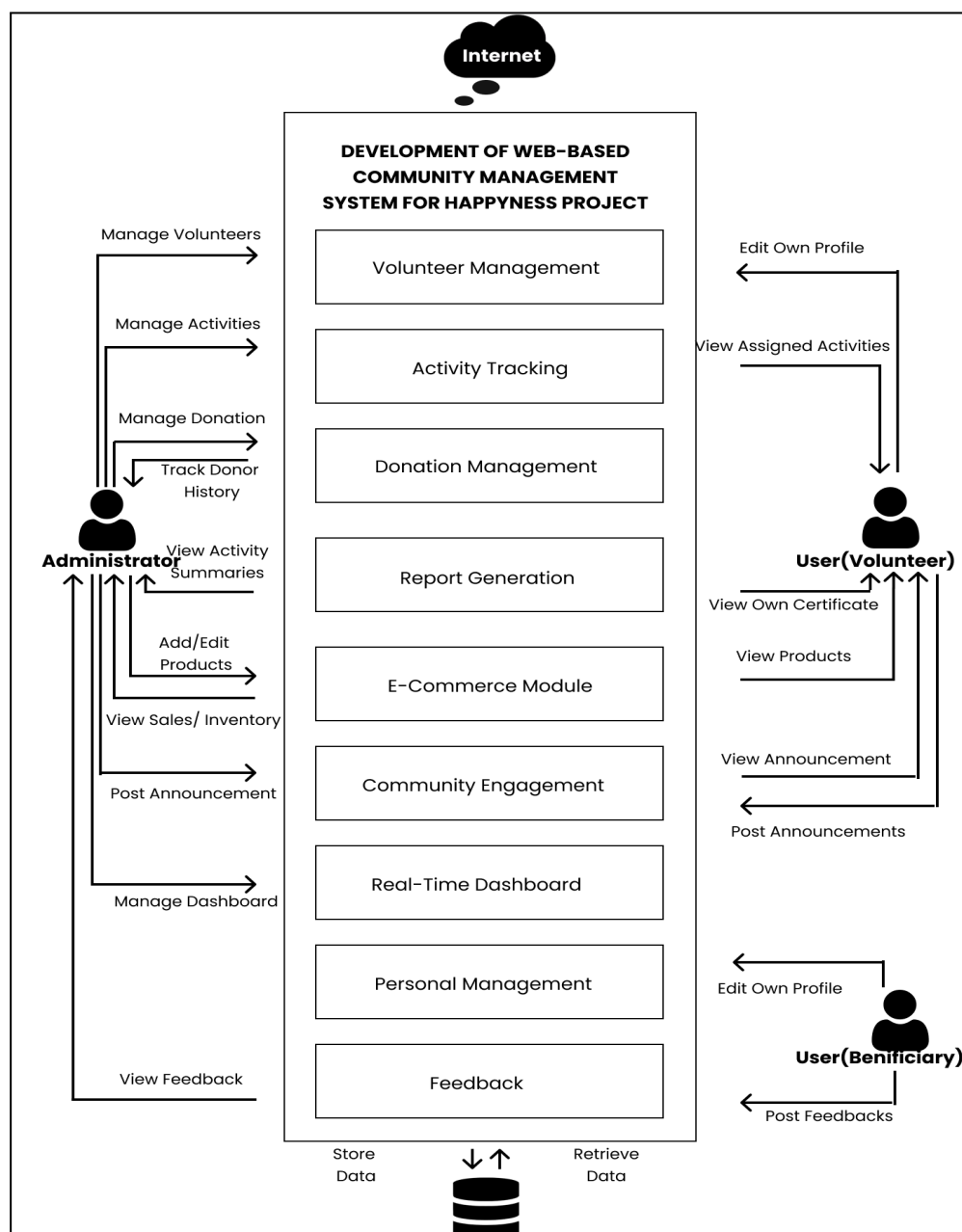
The HappYness Project currently relies on a fully manual process for managing community programs and beneficiaries, involving project teams, coordinators, and volunteers. The process includes distributing surveys to assess community needs, reviewing data, planning programs, registering beneficiaries, and preparing materials. After program delivery, feedback is manually collected and recorded for future improvement.



**Figure 2.** Manual Process Flow of the HappYness Project

Figure 2 illustrates the manual workflow of program and beneficiary management for the HappYness Project. The process begins with conducting a community needs assessment to identify the most urgent concerns. Based on the findings, programs are planned and designed, followed by the announcement of

fundraising campaigns to gather resources. Donations are then received and recorded manually, while volunteers are recruited and assigned to assist in various activities. The collected goods are recorded and organized in inventory before being distributed to beneficiaries as part of program implementation. After distribution, expenses are documented, reports are generated, and feedback is collected to evaluate program outcomes. This manual approach is highly dependent on paperwork and human coordination, which often leads to inefficiencies, data inconsistencies, and delays—emphasizing the need for a digital, automated system to streamline operations and improve transparency.





## **Figure 7. System Architecture**

**Volunteer Management.** Administrators oversee volunteer registration, approval, and task assignments for various outreach activities. Volunteers can update their profiles, monitor their assigned programs, and track participation history, ensuring efficient coordination.

**Activity Tracking.** This module monitors the implementation of community activities such as donation drives, planting initiatives, and educational programs. Administrators can create and assign activities, volunteers can record their participation, and beneficiaries can view or register for community-based programs.

**Donation Management.** Handles both monetary and in-kind donations by tracking donor information, transaction details, and donation history. Administrators manage records and ensure transparency, while beneficiaries can acknowledge received aid or request assistance when eligible.

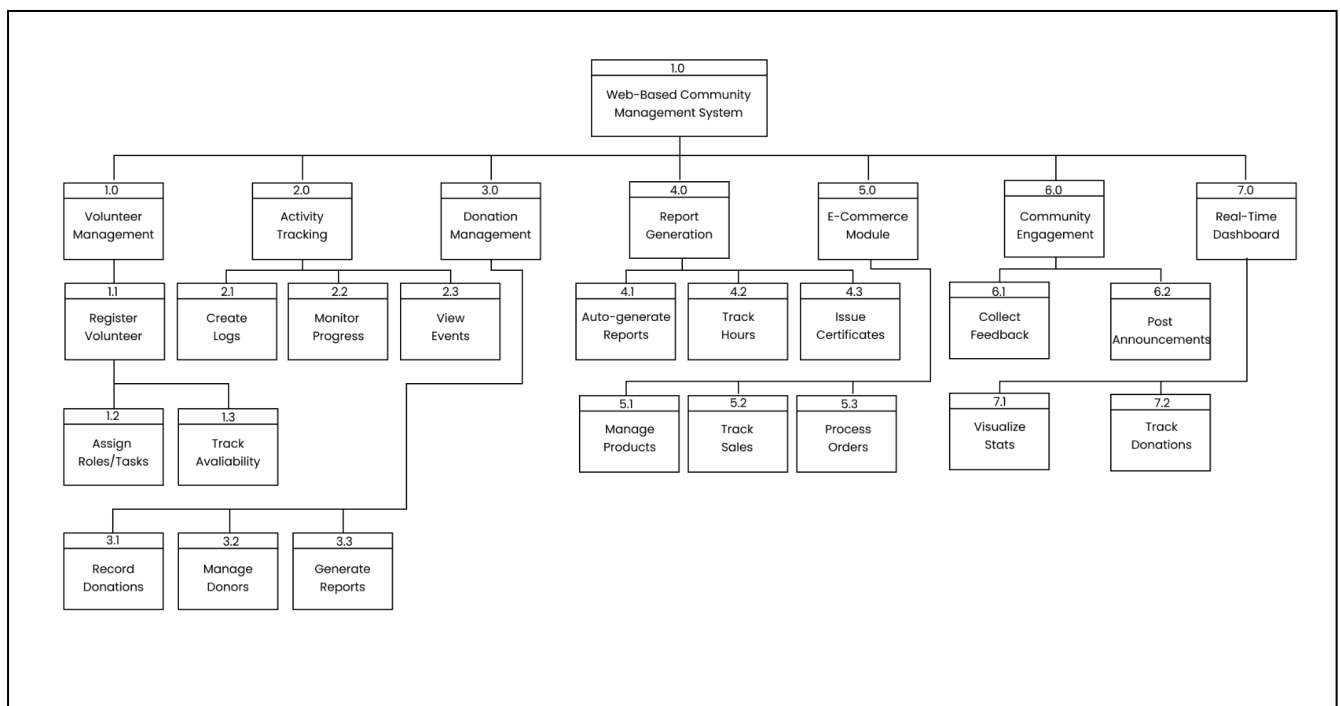
**Beneficiary Management.** A newly added module that allows administrators to register, verify, and manage beneficiary profiles. Beneficiaries can update their information, view distributed goods or services received, and participate in community programs.

**Report Generation.** Automatically compiles data from all modules—volunteer activities, donations, and beneficiary records—into summarized reports. These reports enhance transparency and decision-making for administrators and stakeholders.

**E-Commerce Module.** Facilitates the sale of community-produced goods such as rags, plants, and crafts. Administrators manage product listings and transactions, while beneficiaries and donors can purchase products to support the project.

**Community Engagement.** Provides an interactive platform for communication and feedback. Users—including beneficiaries—can send feedback, view announcements, and stay informed about upcoming initiatives and events.

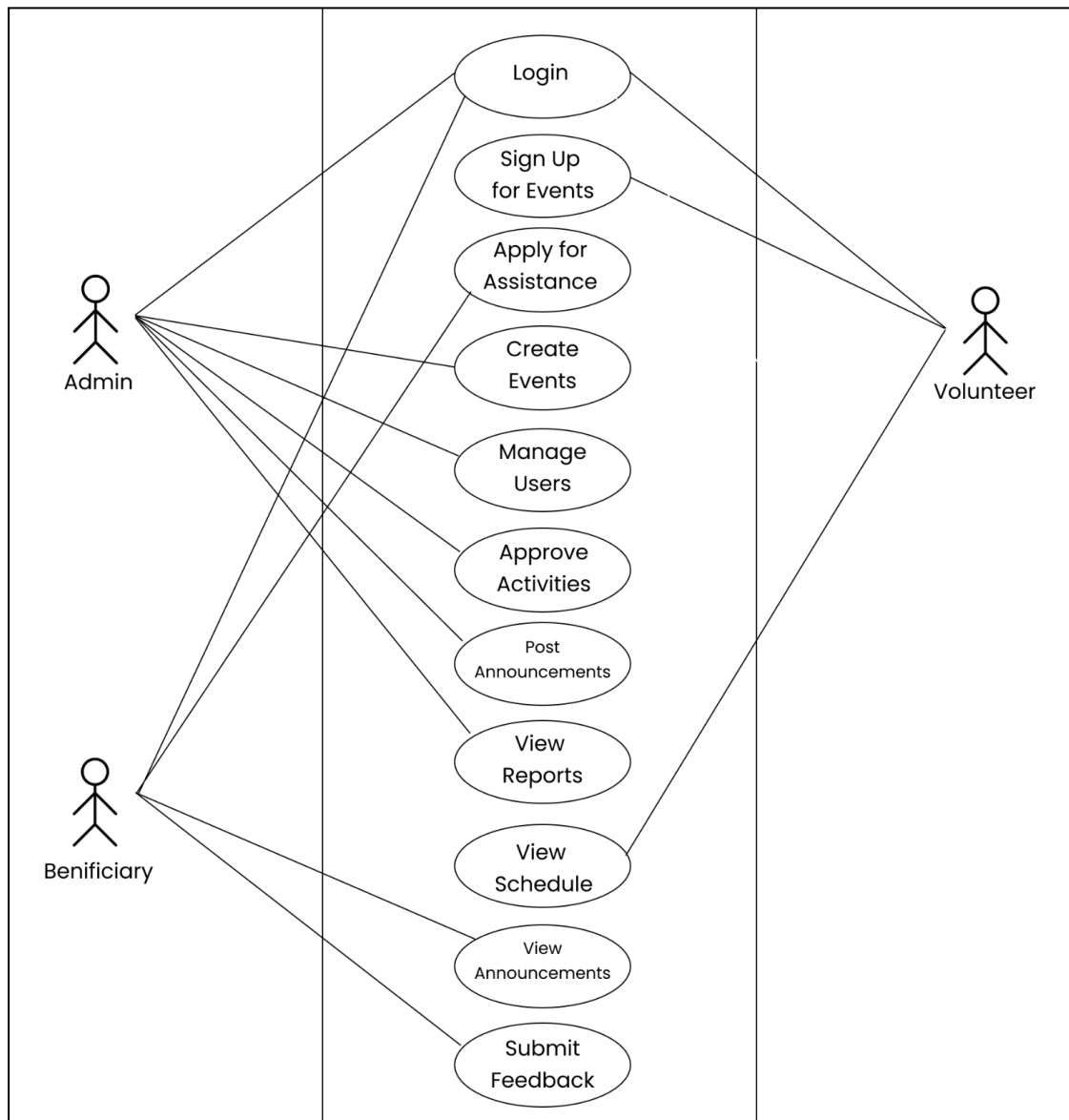
**Real-Time Dashboard.** Displays live data on donations, volunteer participation, and program status. Administrators can easily monitor ongoing activities, track beneficiary engagement, and make data-driven decisions.



**Figure 8.** HIPO Diagram

Figure 8 shows the HIPO diagram for the Web-Based Community Management System of the Happiness Project. It includes modules for Volunteer Management, Activity Tracking, Event Engagement, Community Feedback,

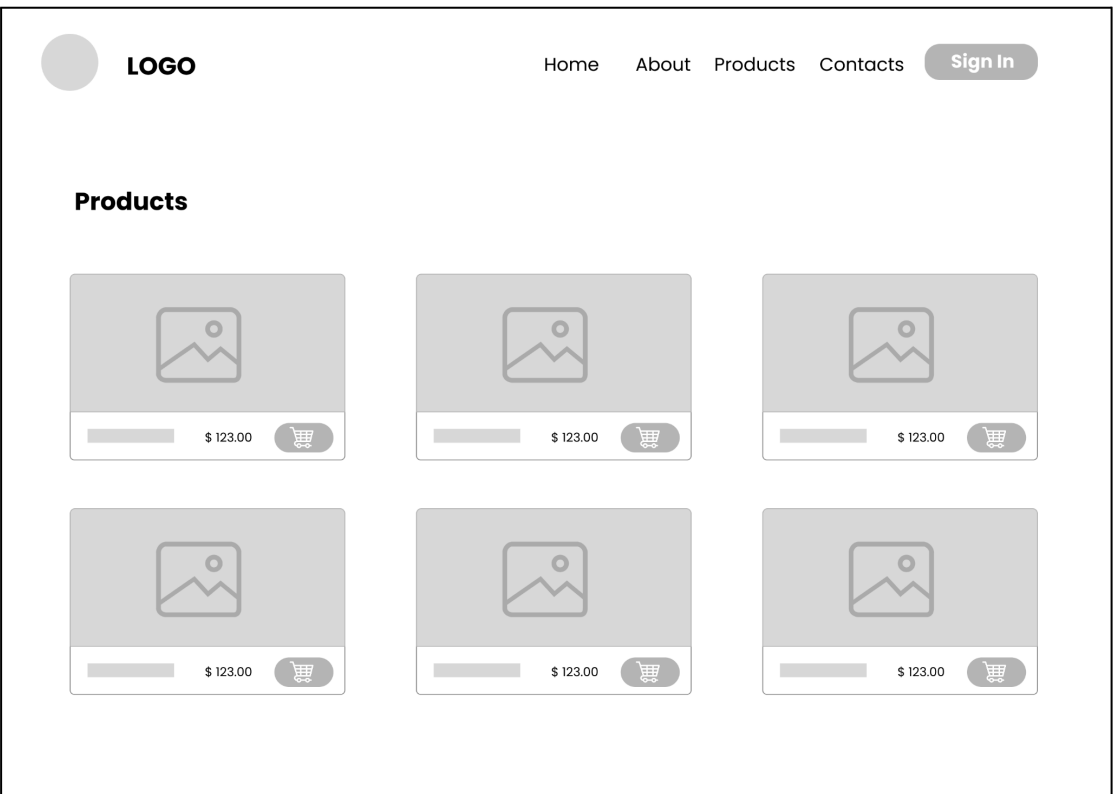
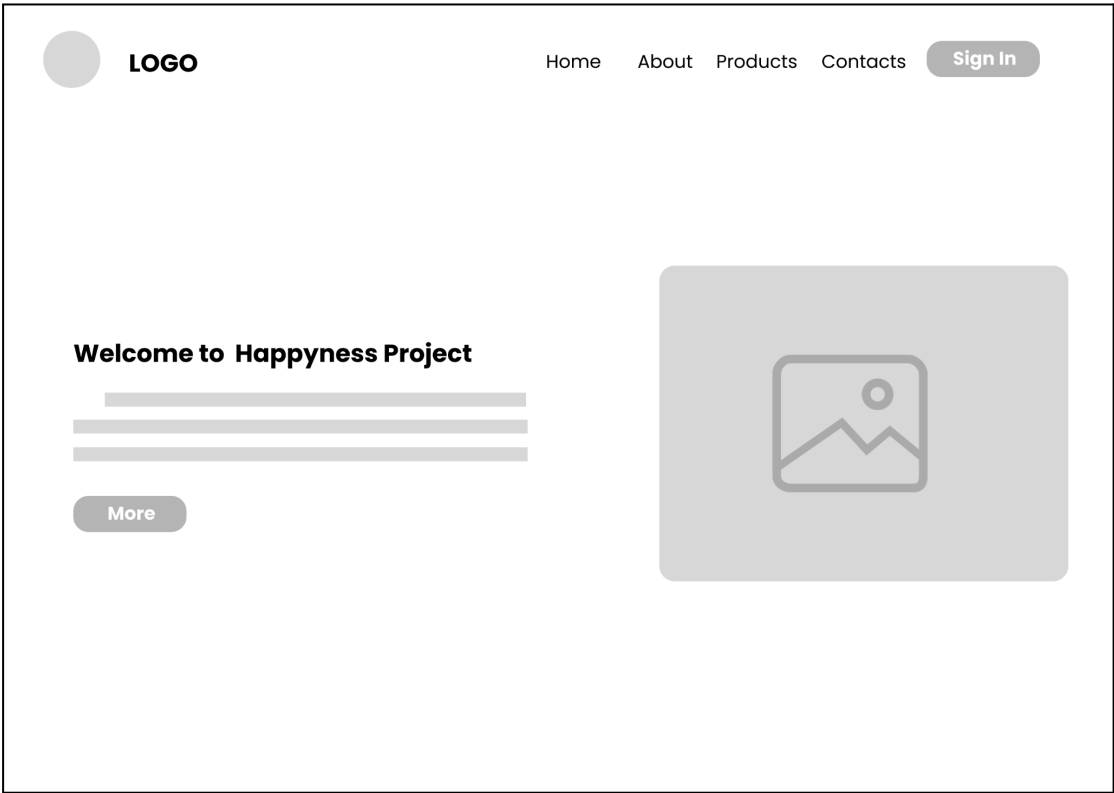
Announcements, and Donation Handling. Each module handles specific functions like volunteer registration, project monitoring, event organization, feedback collection, announcements, and donation management, ensuring efficient operation and coordination.

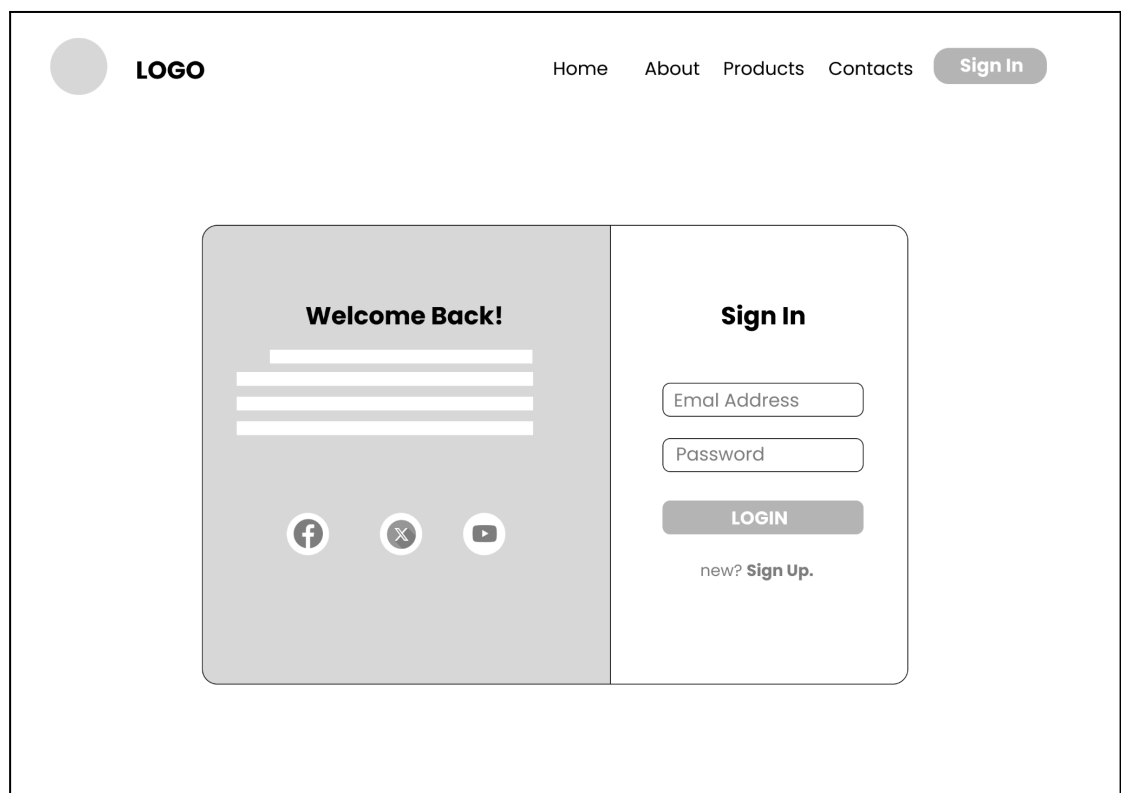
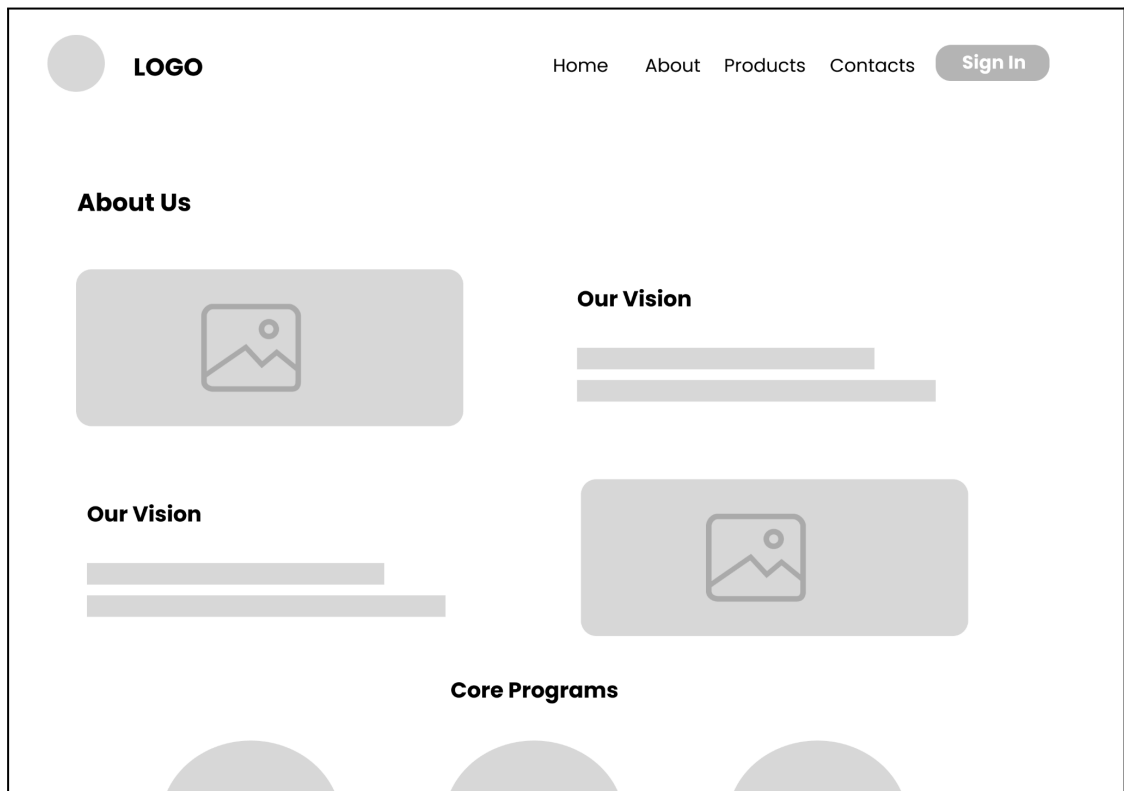


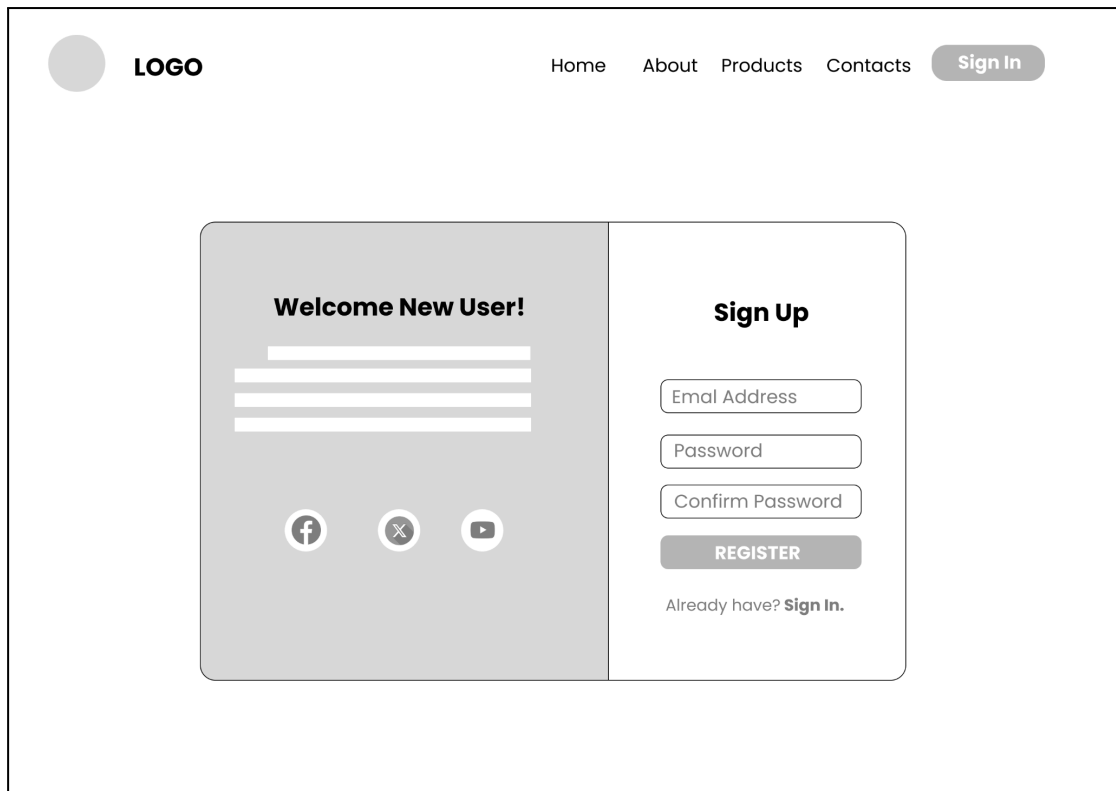
**Figure 9.** Use Case Diagram of the Web-Based Management System

Figure 9 shows the use case diagram for the Web-Based Community Management System, highlighting three user roles—Admin, Volunteer, and

Beneficiary. All users can log in, view announcements and schedules, and submit feedback. Admins have extended privileges to create events, manage users, approve activities, post announcements, and view reports. Volunteers can sign up for events, while beneficiaries can apply for assistance. This diagram outlines how each user interacts with the system to facilitate community engagement.



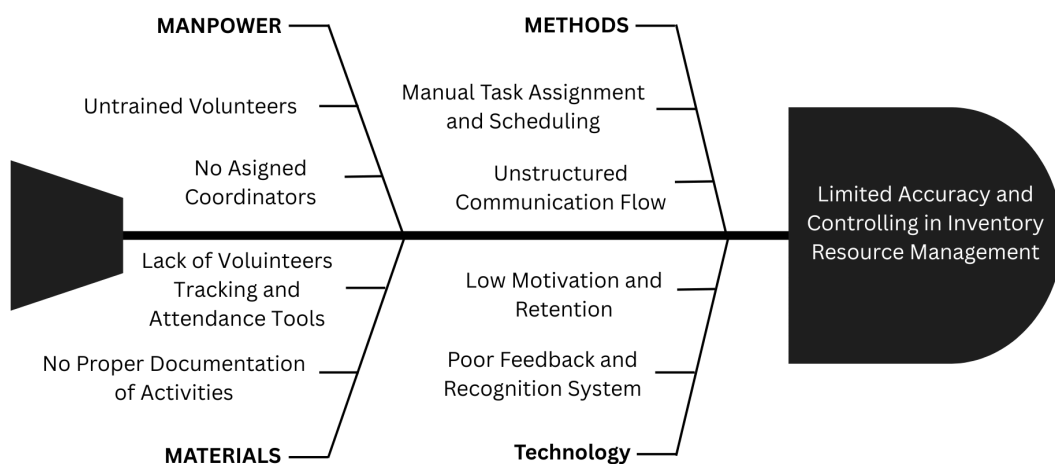




**Figure 10.** System Wireframe

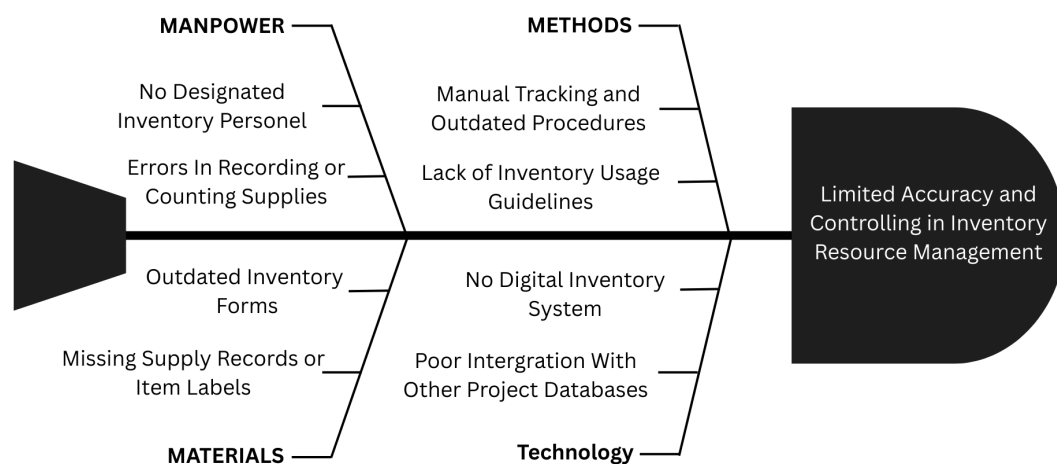
## Requirements Analysis

The researcher collected comprehensive information about the study's needs and requirements during the requirements analysis phase. This was done by conducting interviews



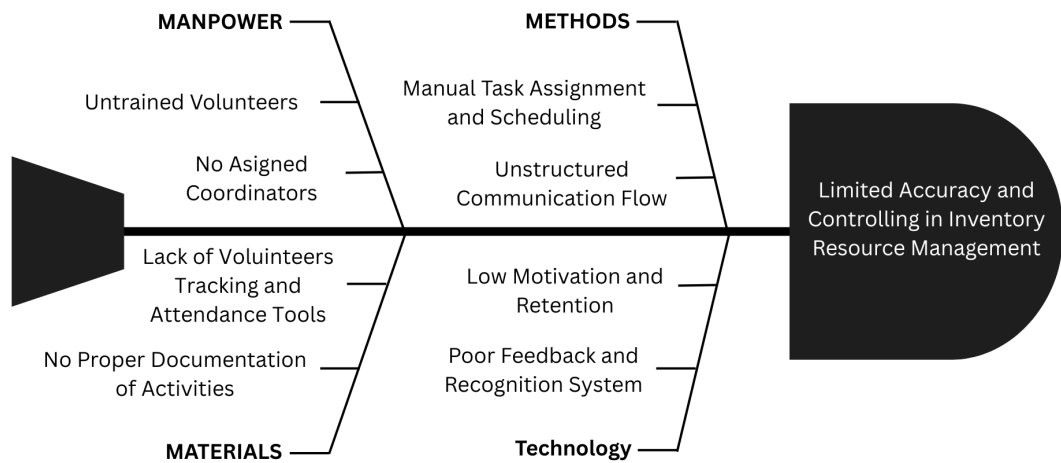
**Figure 11.** Fishbone Diagram of Issues in Donor and Beneficiary Management.

The Manpower includes limited training in donor handling and inexperienced volunteers. Methods involve manual tracking of donations and unclear reporting processes. Materials include a lack of standardized donor forms and inconsistent donation records, while Technology refers to the absence of a centralized database and weak data backup systems.



**Figure 12.** Fishbone Diagram of Problems in Volunteer Management.

The Manpower Includes untrained volunteers and the lack of assigned coordinators. Methods involve manual task scheduling and poor communication flow. Materials include no proper documentation and a lack of tracking tools, while Technology refers to low motivation and weak feedback systems that affect volunteer engagement.



**Figure 13.** Fishbone Diagram of Factors Affecting Inventory and Resource Management.

The Manpower includes no assigned inventory staff and errors in recording or counting supplies. Methods involve manual tracking, outdated procedures, and lack of clear inventory guidelines. Materials include outdated forms and missing item records, while Technology refers to the absence of a digital system and poor integration with other project databases.

**Requirement Analysis.** To gather all the essential information about the stakeholders' specific requirements for the project, the researchers will conduct thorough interviews during the requirement analysis phase. This participatory approach guarantees a comprehensive understanding of the stakeholders' needs, expectations, and desired outcomes, forming a solid foundation for the subsequent stages of the project development lifecycle.



The Web-Based Community Management System for the Happiness Project streamlines community initiatives by assigning users into Admins, Donors, and Volunteers. After a secure login, admins oversee inventory management and system security, donors contribute through an e-commerce store, and volunteers manage tasks via a dedicated dashboard.

Admins track donated goods and ensure data security, while donors support projects by purchasing handmade products or making direct contributions. Volunteers coordinate schedules and assignments through a task management module. The system generates real-time reports, allowing admins to monitor donations, track inventory, and assess project impact. Donors receive updates on their contributions, and volunteers get notifications for their tasks. This structured approach ensures efficiency, transparency, and collaboration in managing community-driven efforts.

**Requirement Documentation**

Based on the analysis of the project's objectives and discussions with stakeholders, the following features have been identified for the Web-Based Community Management System for the HappyNess Project:

Table 1. Features of the Proposed System

| Main Features        | Detailed Features                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Volunteer Management | <div>1. Allows volunteers to register and maintain personal profiles, including assigned tasks and availability schedules.</div> <div>2. Enables administrators to assign specific roles and duties to volunteers, particularly social work students (e.g., teaching, rag-making, planting).</div> <div>3. Tracks OJT hours, task completion, and performance evaluations, with automated updates and report generation.</div> |

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|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Activity Management                | <ol style="list-style-type: none"><li>1. Provides guides and tutorials for activities like rag-making, planting, and teaching kids.</li><li>2. Supports event scheduling with task assignments and notifications for volunteers.</li><li>3. Stores activity data for monitoring and future planning.</li></ol>                                                                                                                                                                                                                                                                                                                                |
| Donation and Outreach Distribution | <ol style="list-style-type: none"><li>1. Maintains a beneficiary database including profiles of enrolled mothers and children.</li><li>2. Automates the scheduling and organization of outreach events with alerts for involved parties.</li><li>3. Tracks donations (monetary and in-kind), including inventory management and distribution logs.</li><li>4. Facilitates volunteer coordination for donation and outreach activities.</li><li>5. Documents impact through data and media uploads (e.g., goods delivered, event photos).</li><li>6. Collects feedback from beneficiaries regarding received donations and services.</li></ol> |
| Community Engagement Tools         | <ol style="list-style-type: none"><li>1. Provides a built-in messaging system for secure communication between administrators, volunteers, and members.</li><li>2. Sends announcements for upcoming events, program launches, and important updates.</li><li>3. Supports community forums for open discussion and experience-sharing among members.</li><li>4. Offers special discussion sections for mothers and children to exchange support and advice.</li><li>5. Publishes success stories and testimonials to promote engagement and highlight impact.</li></ol>                                                                        |

|                            |                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-Commerce and Fundraising | <ol style="list-style-type: none"> <li>1. Hosts an online store showcasing handmade products created in community projects.</li> <li>2. Supports multiple online payment methods including bank transfers and donation slips.</li> <li>3. Provides space for promotional advertisements to generate additional program revenue.</li> </ol>                                            |
| Library System             | <ol style="list-style-type: none"> <li>1. Enables online borrowing and returning of books, with digital logging for transactions.</li> <li>2. Tracks book availability, due dates, and sends overdue reminders to users.</li> </ol>                                                                                                                                                   |
| Event Calendar             | <ol style="list-style-type: none"> <li>1. Displays all scheduled community events, activities, and training sessions in a centralized view.</li> <li>2. Sends automated reminders and notifications to participants about upcoming events.</li> </ol>                                                                                                                                 |
| Reporting and Analytics    | <ol style="list-style-type: none"> <li>1. Generates detailed reports on volunteer activity, OJT tracking, and evaluation results.</li> <li>2. Monitors key impact metrics, including children educated, rags produced, trees planted, and distributed donations.</li> <li>3. Provides donor transparency through visualized reports on fund utilization and program reach.</li> </ol> |

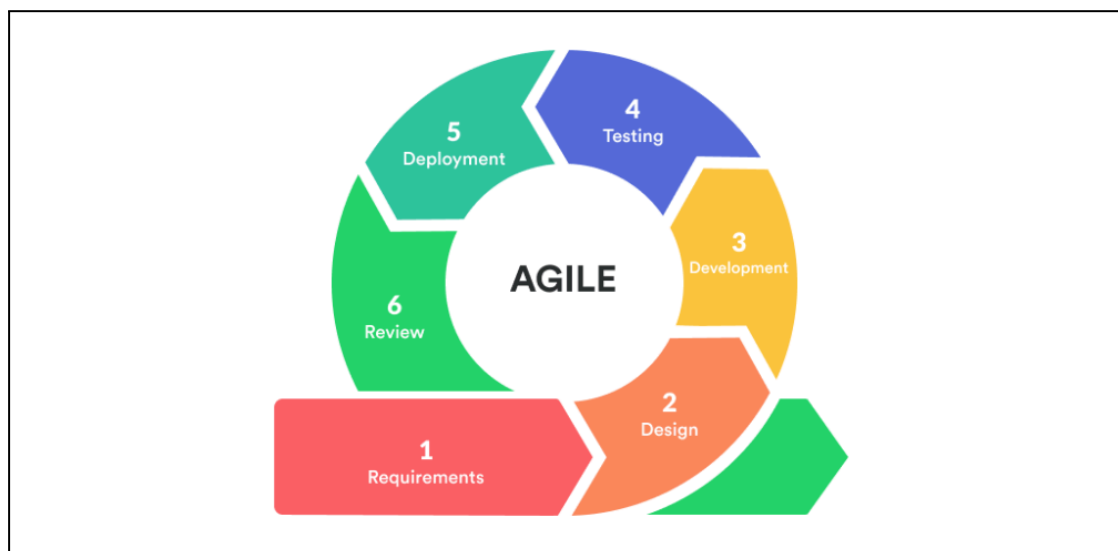
Table 2. Non-Functional Requirements of the System

| Quality Attributes | Description                                                                                    |
|--------------------|------------------------------------------------------------------------------------------------|
| Time Behavior      | Pages and features shall load within 5 seconds for optimal user experience.                    |
| Usability          | The system shall offer a consistent and intuitive interface across all modules.                |
| Accessibility      | Supports multiple languages and includes features for color-blind and visually impaired users. |

|             |                                                                                          |
|-------------|------------------------------------------------------------------------------------------|
| Security    | Inactive users shall be logged out automatically; all sensitive data shall be encrypted. |
| Scalability | Capable of supporting up to 1,000 concurrent users without performance issues.           |
| Compliance  | Adheres to WCAG 2.1 standards to ensure accessibility for all users.                     |

## System Development

The development of the Web-Based Community Management System for the HappyNess Project followed the Agile model, a flexible and iterative software development methodology. This approach breaks down the development process into manageable cycles that include Requirements, Design, Development, Testing, Deployment, and Review (see Figure 15: Agile Development Model). Each cycle allows the team to improve the system continuously, based on user feedback and changing project needs. By gradually adding new features and refining existing ones, the system evolves toward a fully functional solution that supports the goals of the HappyNess Project.



**Figure 15.** Agile Development Model

**Requirements.** In the requirements phase, the researchers began by identifying the key challenges faced by the HappYness Project in its manual operations, such as disorganized volunteer tracking, donor management, and activity monitoring. To gain deeper insights, interviews and observations were conducted across multiple project areas including Bacoar, Quiapo, and Binangonan. A fishbone diagram was created to visualize the root causes of operational inefficiencies. Based on these findings, the research team outlined the system's primary objectives: streamline volunteer coordination, track project activities, improve donation processes, and enable efficient reporting. These goals formed the basis for the first development iteration.

**Design.** During the design phase, the research team created initial design concepts, including wireframes (see Figure 10: System Wireframe) and Use Case Diagrams, that reflected the features required by the stakeholders. The system architecture was structured to support different user roles—Admin, Volunteer, and Beneficiary—ensuring that the interface and functionalities aligned with their respective responsibilities. Through stakeholder consultations and collaborative refinements, the system's visual structure and user flows were finalized to ensure usability and purpose-driven design.

**Development.** The development phase involved the actual construction of the system using modern web technologies. The team employed HTML, CSS, and JavaScript for the frontend, while PHP and MySQL were used for the backend development and database management. The system was divided into functional modules: Volunteer Management, Donation Tracking, Activity Monitoring, Inventory Management, E-Commerce, and Reporting. Each module was developed

incrementally, allowing for parallel work on different features while maintaining alignment with the initial goals defined during the requirements phase.

**Testing.** Once a functional version of the system was developed, it underwent rigorous testing. Each feature was tested for functionality, data accuracy, and user experience. Bugs and logical issues identified during this phase were documented and resolved promptly. The goal of this phase was to ensure the stability and reliability of the system, and to verify that it fulfilled the needs outlined by the HappYness Project stakeholders.

**Deployment.** After successful testing, the system was deployed in a controlled environment for real-world use. This involved installing the web application on a live server and configuring it for user access. Admin users were trained on how to navigate and manage the platform, and initial feedback was collected for evaluation. The deployment allowed stakeholders to begin using the core features while additional improvements were planned for future iterations.

**Review.** The final phase involved reviewing the system's performance and gathering feedback from users and project beneficiaries. The review process helped identify areas for improvement and additional feature requests, ensuring that future updates would better support the evolving needs of the HappYness Project. Based on this feedback, the next iteration of the Agile cycle was planned, continuing the system's development in alignment with real-world application.

